Unit 3 - Calculus
3.3 Integrals

Graham West

MSDS 520: Math and Stats Foundations for Data Science
Meharry Medical College

Outline

1. Why Integrals?
2. Distance from Velocity
3. Riemann Sums
4. The Definite Integral
5. Integrals from Data
6. Antiderivatives and the Fundamental Theorem
7. Applications of Integrals
8. 2D Integrals
9. Review

Why Integrals?

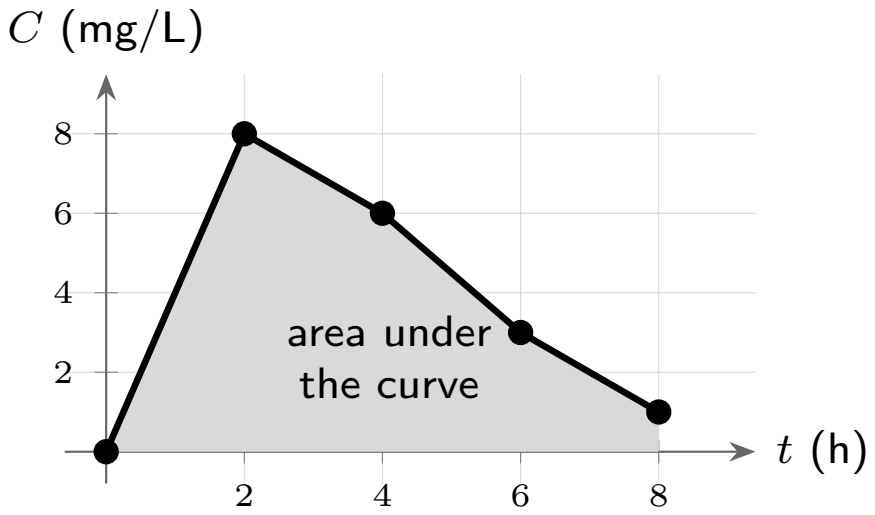
A Problem to Motivate Us

Unit 3-1 went one way

position $x(t)$ $\xrightarrow{\text{derivative}}$ velocity $v(t)$

Today runs it backward

- Only the **speedometer** $v(t)$: how **far** did the car go?
- Only a patient's **blood draws**: how much drug did the body **see**?



Five blood draws after one dose.

The one idea behind both questions

A **rate** times a **time** is an **amount**. When the rate keeps changing, the amount is the **area under the rate's graph**. That area is an **integral**.

Where This Shows Up in Data Science

Integrals underneath the tools

- **Pharmacokinetics.** The **AUC** (area under the concentration-time curve) is the standard measure of total drug exposure. It sets doses, and a generic drug is approved only if its AUC matches the brand's.
- **Probability.** For a continuous measurement (blood pressure, a lab value), the probability of landing in a range is the **area under a density curve**. You will meet this in your statistics courses.
- **Evaluating classifiers.** A model's **ROC AUC** is an area under a curve: about 0.5 for a coin flip, 1 for a perfect model.
- **Totals from rates.** Energy from a power meter (kW $\to$ kWh), fluid from an IV pump's rate (mL/h $\to$ mL), new cases from a daily case rate.

What We Will Build

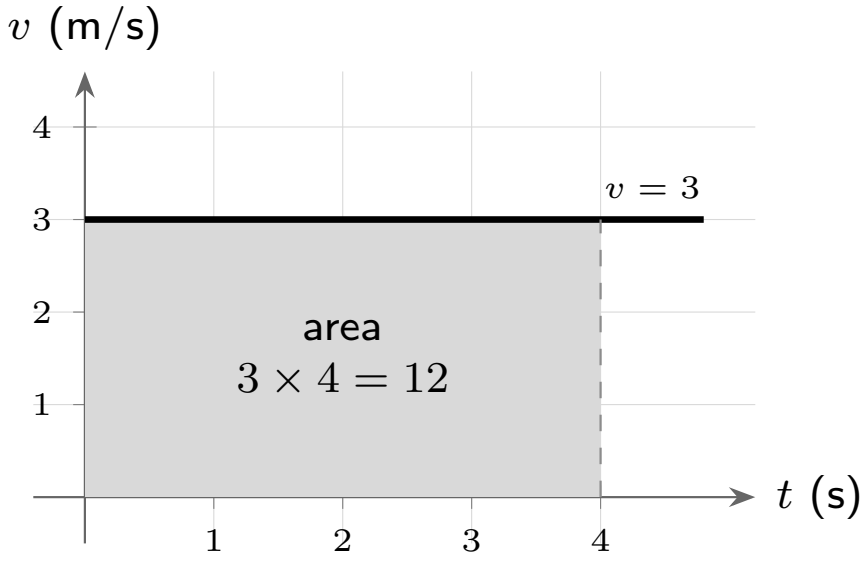
The plan

1. **Distance from velocity**: distance is the area under the velocity graph.
2. **Riemann sums**: approximate any area with rectangles (left, right, midpoint).
3. The **definite integral**: the limit of those sums, its notation, signed area, units.
4. **Integrals from data**: the trapezoid rule, AUC from blood draws, and Python.
5. **Antiderivatives and the Fundamental Theorem**: integration undoes differentiation, which gives exact areas with no rectangles at all.
6. **Applications**: average value, and continuous probability distributions.
7. **2D integrals**: boxes under a surface, volume, and the double integral.

Notebooks for this lecture: `U3-8_Integrals-1` to `-5`; `DiscreteVsContinuous`.

Distance from Velocity

Constant Velocity: Distance Is a Rectangle



A car at a steady 3 m/s for 4 s

- Distance = velocity $\times$ time:

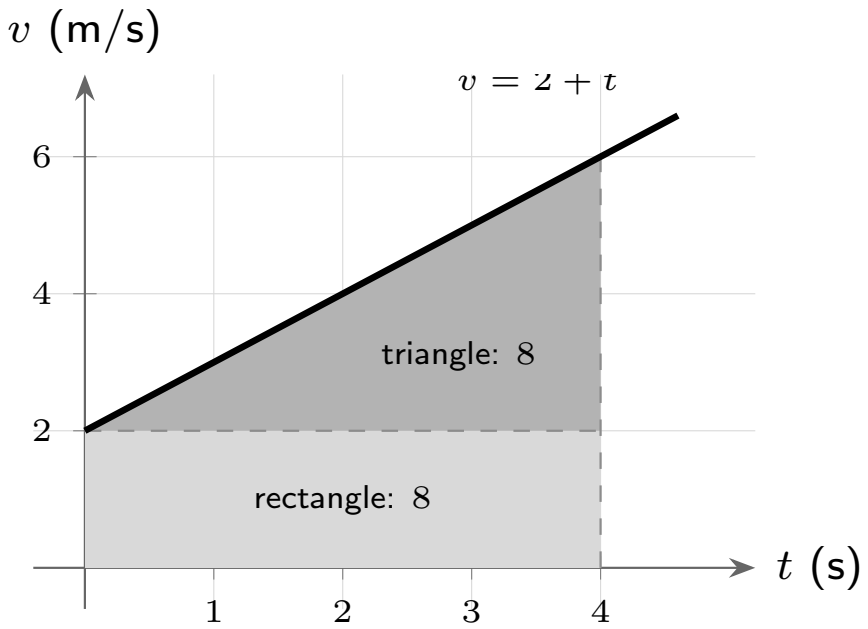
$$d = 3 \cdot 4 = 12 \text{ m}.$$

- On the graph, that product is the **rectangle** under the line: height 3, width 4.
- Units agree: $\frac{\text{m}}{\text{s}} \cdot \text{s} = \text{m}$.

Key idea

Distance traveled = **area under the velocity graph**. Same for an IV pump: 100 mL/h for 3 h is a rectangle of area 300 mL.

Speeding Up Steadily: Rectangle Plus Triangle



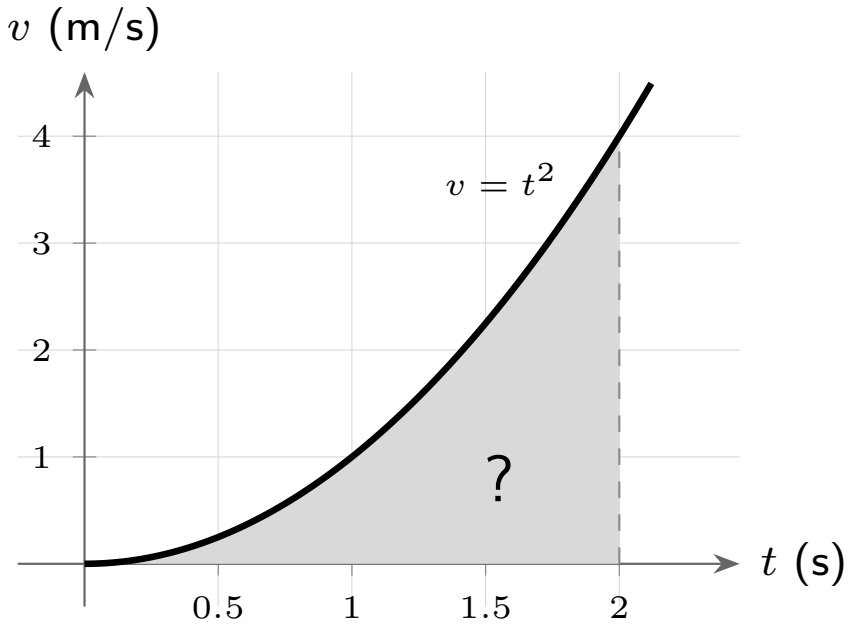
$v(t) = 2 + t$ for $t = 0$ to 4 s

- No single velocity, so no single product.
- But the area is still easy geometry:
  - rectangle: $2 \times 4 = 8$
  - triangle: $\frac{1}{2} \times 4 \times 4 = 8$
- Distance: $8 + 8 = 16$ m.

Check

The velocity climbs evenly from 2 to 6, so its average is 4 m/s, and $4 \times 4 = 16$ m. The area idea still works when velocity changes.

A Curved Velocity: Geometry Runs Out



A car pulling away: $v(t) = t^2$

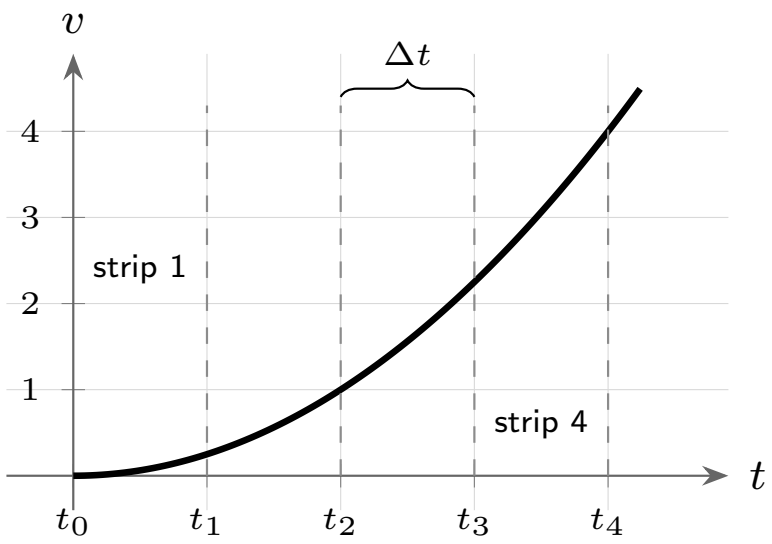
- How far in the first 2 seconds?
- The answer is still the area under the graph.
- But the top edge is **curved**: no rectangle, triangle, or circle formula fits.

The calculus move: solve a simpler problem

- Rectangles we **can** do. Cover the region with thin ones and add them up.
- Make them thinner until the error disappears.
- $v = t^2$ on $[0, 2]$ is the running example for the rest of the lecture.

Riemann Sums

Slice the Interval into Strips



Step 1: cut $[a, b]$ into n strips

- Equal widths: $\Delta t = \frac{b-a}{n}$.
- Cut points: $t_i = a + i\,\Delta t$, from $t_0 = a$ to $t_n = b$.
- Car: $[0, 2]$ with $n = 4$, so $\Delta t = 0.5$; cut points $0, 0.5, 1, 1.5, 2$.

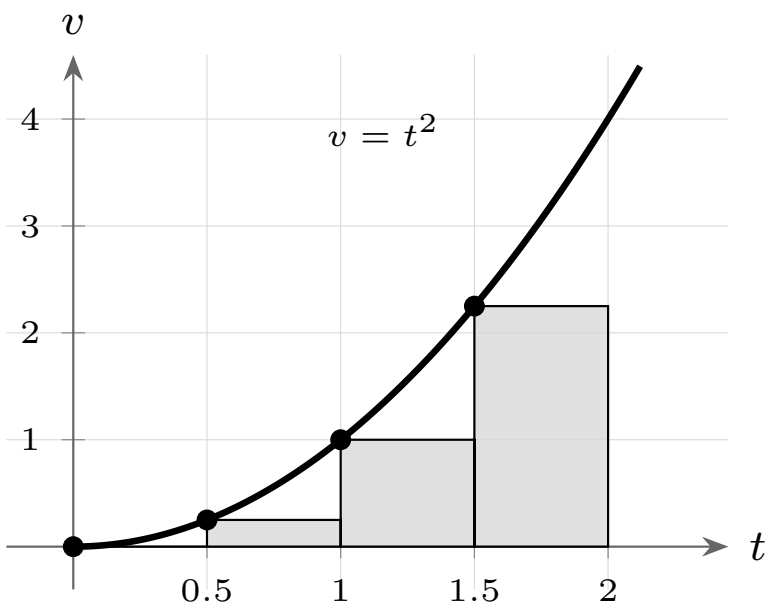
Step 2: one rectangle per strip

Width Δt; height = the function's value at **one chosen point** in the strip: its left end, right end, or middle.

Notebook `U3-8_Integrals-1_RiemannSums`, §1

The Left Endpoint Rule

heights from the **left** ends



Heights at the left ends

strip	**left end**	**height** t^2
1	0	0
2	0.5	0.25
3	1	1
4	1.5	2.25
	sum	3.5

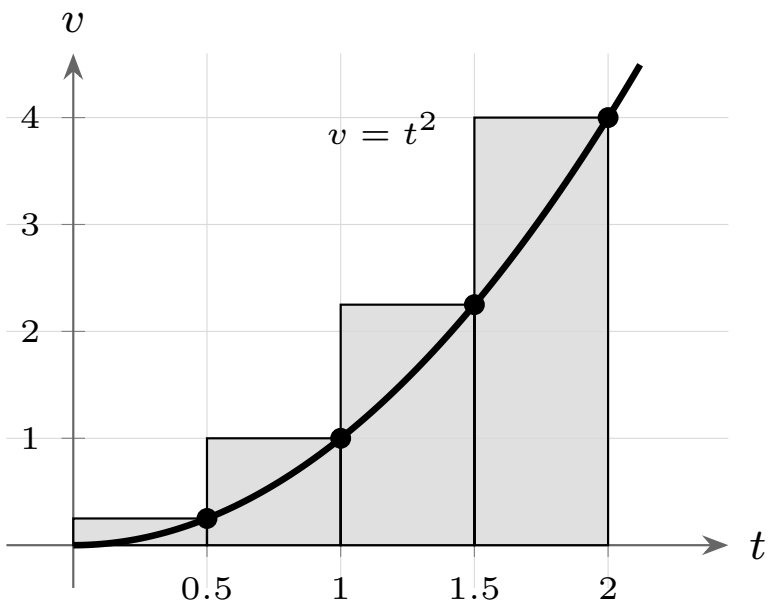
Area $\approx 3.5 \times 0.5 = \mathbf{1.75}$ m.

Watch out: too low

$v = t^2$ is **increasing**, so each strip's left end is its lowest point. Every rectangle sits **under** the curve, and 1.75 is an **underestimate**.

The Right Endpoint Rule

heights from the **right** ends



Heights at the right ends

strip	right end	height t^2
1	0.5	0.25
2	1	1
3	1.5	2.25
4	2	4
	sum	7.5

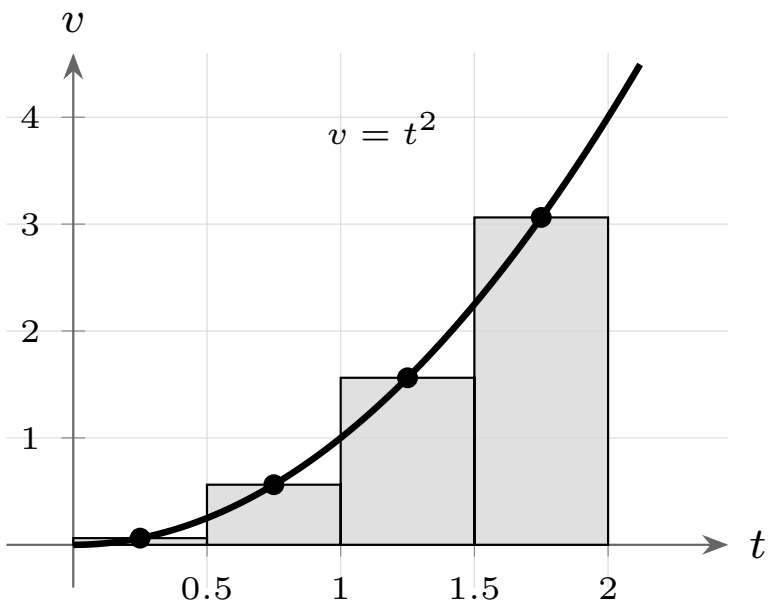
Area $\approx 7.5 \times 0.5 = \mathbf{3.75}$ m.

Watch out: too high

Now each strip uses its **highest** point, so every rectangle pokes **above** the curve: 3.75 is an **overestimate**. The true area is somewhere between 1.75 and 3.75.

The Midpoint Rule

heights from the **middles**



Heights at the middles

strip	**middle**	**height** t^2
1	0.25	0.0625
2	0.75	0.5625
3	1.25	1.5625
4	1.75	3.0625
	sum	5.25

Area $\approx 5.25 \times 0.5 = \mathbf{2.625}$ m.

Why the middle does better

Inside each strip the rectangle is too high on one side and too low on the other. The two errors **nearly cancel**, so the same 4 rectangles land much closer to the truth.

The Riemann Sum

Definition (named for Bernhard Riemann, 19th century)

For f on $[a, b]$ cut into n strips of width $\Delta x = \frac{b-a}{n}$, pick a point x_i^* in strip i. The **Riemann sum** is

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x \;=\; f(x_1^*)\,\Delta x + f(x_2^*)\,\Delta x + \cdots + f(x_n^*)\,\Delta x.$$

Reading it

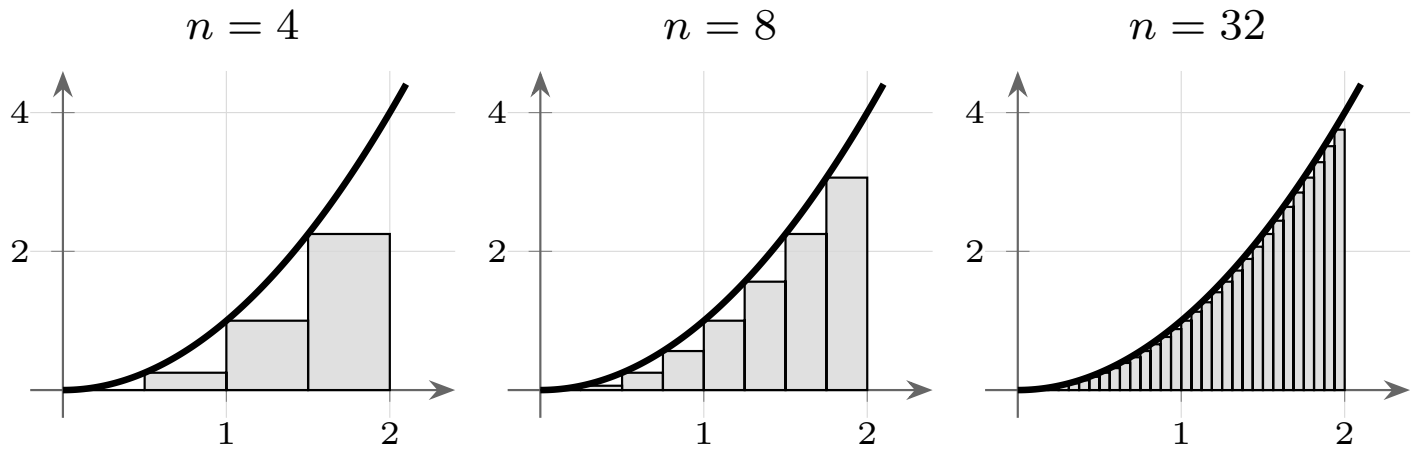
- $\sum_{i=1}^{n}$: add up the strips (Unit 1-4).
- $f(x_i^*)$: **height** of rectangle i.
- Δx: its **width**.
- Strip i runs from x_{i-1} to x_i.

The three rules

rule	**chosen point** x_i^*
left	x_{i-1}
right	x_i
midpoint	$\frac{1}{2}(x_{i-1} + x_i)$

Notebook `U3-8_Integrals-1_RiemannSums`, §2

More Rectangles, Closer Answers



Left rule for $v = t^2$ on $[0, 2]$: the gaps above the rectangles shrink as n grows.

Area under $v = t^2$ on $[0, 2]$

n	**left**	**right**	**midpoint**
4	1.7500	3.7500	2.6250
8	2.1875	3.1875	2.6563
16	2.4219	2.9219	2.6641
100	2.6268	2.7068	2.6666

What the table says

- All three close in on **one number**, about 2.667.
- Left and right squeeze it from below and above.
- Midpoint gets there fastest.

Notebook `U3-8_Integrals-1_RiemannSums`, §3 and §4

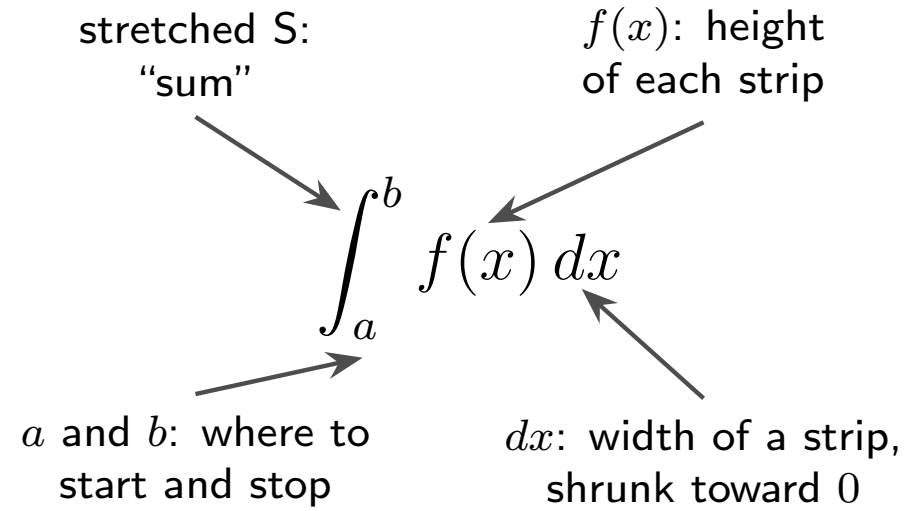
The Definite Integral

From Sums to the Integral

Let the rectangles get infinitely thin

Take more and more strips, $n \to \infty$, so the width $\Delta x \to 0$. The Riemann sums settle on one number: the **definite integral** of f from a to b,

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x \;=\; \int_a^b f(x)\,dx.$$

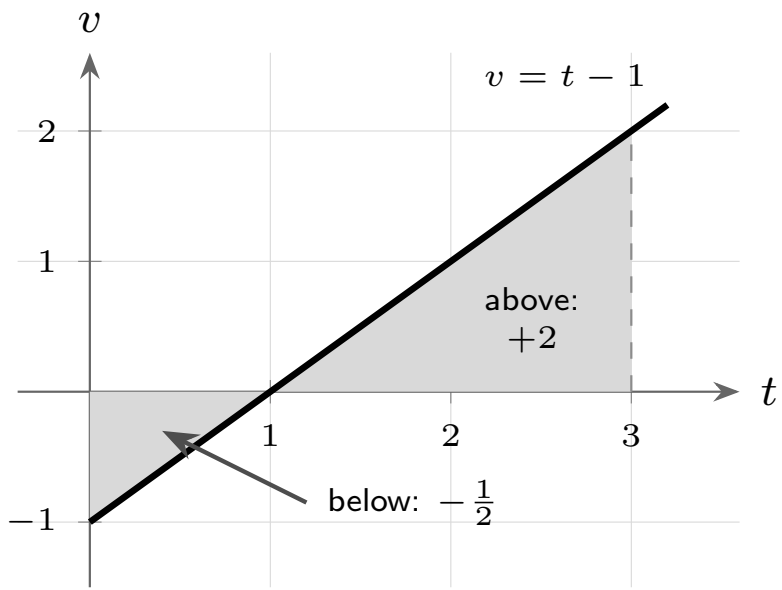


Sum and integral, side by side

	sum	**integral**
add up	$\sum$	$\int$
height	$f(x_i^*)$	$f(x)$
width	Δx	dx
range	$i = 1$ to n	$x = a$ to b

a and b are the **limits of integration**. The letter after d names the variable: $\int_0^2 t^2\,dt$ adds up along t.

Signed Area: Below the Axis Counts Negative



The rule

- Where $f(x) < 0$, the heights $f(x_i^*)$ are negative, so those rectangles **subtract**.
- $\int_0^3 (t-1)\,dt = 2 - \frac{1}{2} = \frac{3}{2}$.

What it means: net, not total

- $v < 0$ means the car is **backing up**. For $t = 0$ to 1 it backs up $\frac{1}{2}$ m, then drives forward 2 m.
- The integral gives where it **ends up**: $\frac{3}{2}$ m ahead (net displacement).
- Total ground covered is $\frac{1}{2} + 2 = \frac{5}{2}$ m. That needs the areas added **without** signs.

Units: What an Integral Measures

Units of $\int_a^b f(x)\,dx$ = (units of f) $\times$ (units of x)

Each rectangle is a height times a width, so the total carries both units.

Rates in, totals out

f **(height)**	x **(width)**	**integral**	**meaning**
velocity, m/s	time, s	m	distance
IV rate, mL/h	time, h	mL	volume given
drug level, mg/L	time, h	mg·h/L	AUC, exposure
power, kW	time, h	kWh	energy used
new cases per day	time, days	cases	total cases

Rule of thumb

If f is a **rate** (something per unit of x), the integral is the **amount** that piled up.

Exercise: Integrals from Geometry

Try it

Sketch each graph and find the integral with area formulas (no rectangles needed).

1. $\int_0^5 2\,dx$
2. $\int_0^4 x\,dx$
3. $\int_{-2}^2 x\,dx$
4. $\int_0^2 (x-2)\,dx$
5. An IV pump runs at $f(x)$ mL/h, with x in hours. What are the units of $\int_0^6 f(x)\,dx$, and what does it measure?

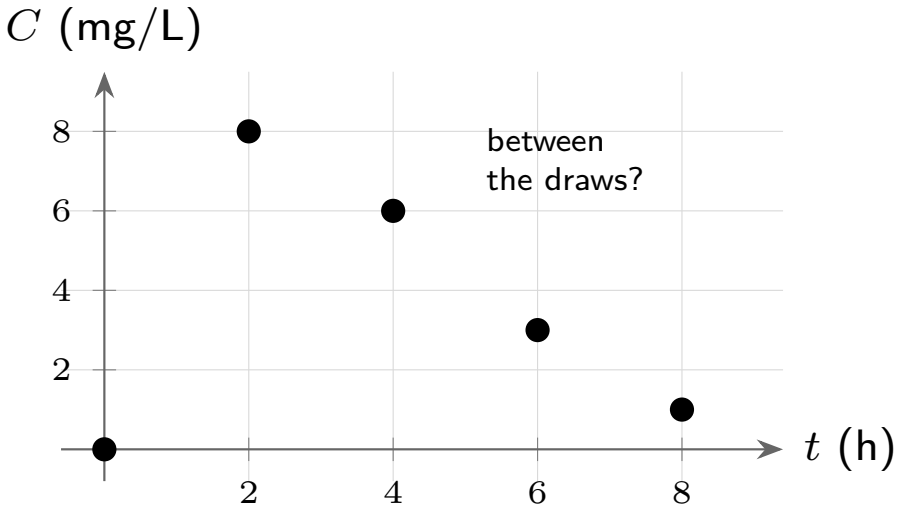
Exercise: Solution

Answers

1. A rectangle, height 2, width 5: area $= 10$.
2. A triangle from $(0, 0)$ to $(4, 4)$: $\frac{1}{2} \times 4 \times 4 = 8$.
3. Two triangles of area $\frac{1}{2} \times 2 \times 2 = 2$: one below the axis (from -2 to 0), one above (from 0 to 2). Signed: $-2 + 2 = 0$. The area is not zero, but the **net** is.
4. The line runs from -2 up to 0, entirely **below** the axis: a triangle of area $\frac{1}{2} \times 2 \times 2 = 2$, counted negative. Integral $= -2$.
5. (mL/h) $\times$ (h) $=$ **mL**: the total volume the patient received in the first 6 hours.

Integrals from Data

When All You Have Is Measurements



Blood draws after one oral dose

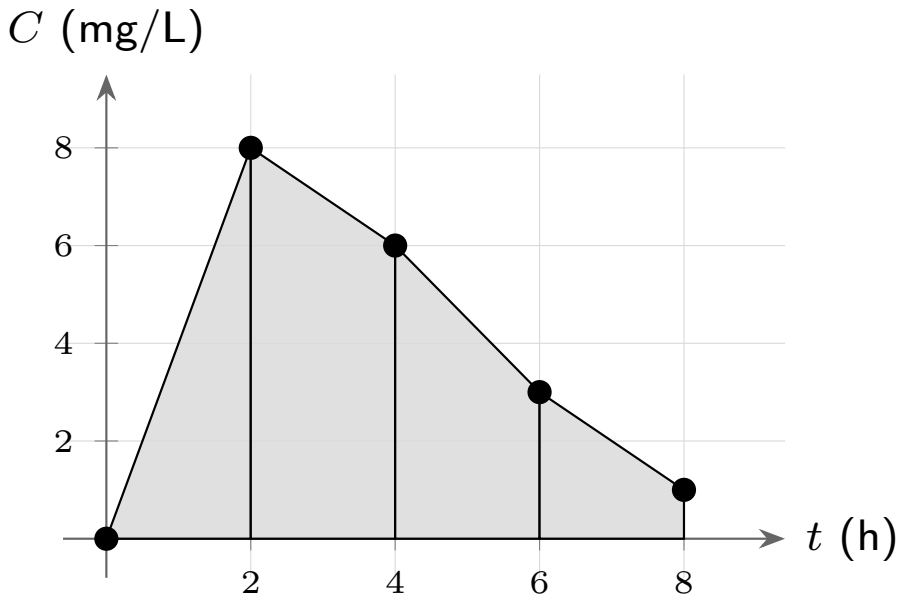
t (h)	0	2	4	6	8
C (mg/L)	0	8	6	3	1

What changes with data

- No formula: we know C only at the draw times.
- Δt is **fixed by the schedule** $(2$ h$)$: the strips cannot get thinner.
- No midpoint rule: nobody measured at $t = 1, 3, 5, 7$.
- Left and right still work: $(0 + 8 + 6 + 3) \times 2 = 34$, $(8 + 6 + 3 + 1) \times 2 = 36$.

Notebook `U3-8_Integrals-3_DataPoints`, §1 and §2

The Trapezoid Rule



Join neighbors with straight lines

- Each strip becomes a **trapezoid**: width Δt, average height $\frac{y_i + y_{i+1}}{2}$.
- Strip area: $\dfrac{y_i + y_{i+1}}{2}\,\Delta t$.
- It is exactly the **average** of the left and right rules.

The trapezoid rule

$$T = \Delta t \left(\tfrac{1}{2}y_0 + y_1 + y_2 + \cdots + y_{n-1} + \tfrac{1}{2}y_n\right)$$

- Uneven draw times (common in real studies) are fine: give each strip its own width.

Worked: AUC from Blood Draws

Strip by strip, $\Delta t = 2$ h

strip	average height	area
0 to 2	$\frac{0+8}{2} = 4$	$4 \times 2 = 8$
2 to 4	$\frac{8+6}{2} = 7$	$7 \times 2 = 14$
4 to 6	$\frac{6+3}{2} = 4.5$	$4.5 \times 2 = 9$
6 to 8	$\frac{3+1}{2} = 2$	$2 \times 2 = 4$
	total	**35**

Check

- Formula:
  $2\left(\frac{1}{2}\cdot 0 + 8 + 6 + 3 + \frac{1}{2}\cdot 1\right)$
  $= 2 \times 17.5 = 35.$
- Average of left and right:
  $\frac{34+36}{2} = 35.$

Answer

- $\text{AUC}_{0 \text{ to } 8\text{h}} \approx 35$ mg·h/L: the patient's total exposure to the drug over the first 8 hours.
- Real studies also estimate the tail after the last draw, since the drug is not fully gone at 8 h.

Notebook `U3-8_Integrals-3_DataPoints`, §4

Integrals from Data in Python

NumPy's `trapezoid`

```
import numpy as np
t = np.array([0, 2, 4, 6, 8])  # hours
C = np.array([0, 8, 6, 3, 1])  # mg/L

np.trapezoid(C, t)    # trapezoid rule
# 35.0
dt = 2                # left and right rules
C[:-1].sum() * dt, C[1:].sum() * dt
# (34, 36)

# a formula: sample it finely
x = np.linspace(0, 2, 101)   # 100 strips
np.trapezoid(x**2, x)
# 2.6668000000000003
```

Reading the call

- `np.trapezoid(y, x)`: **heights first**, then positions (uneven spacing is fine).
- Formula: `linspace` picks the strips. Data: the schedule does.

Watch out

- `...0003` is floating point (Unit 1-5).
- NumPy before 2.0: `np.trapz`.

Notebook `U3-8_Integrals-4_RealData`, §2, a real power meter

Exercise: Brand vs. Generic

Try it

Two formulations of the same drug, blood drawn every hour (mg/L):

t (h)	0	1	2	3	4
brand	0	4	5	3	2
generic	0	6	4	2	1

1. Use the trapezoid rule to find each formulation's AUC from 0 to 4 h.
2. Compute the ratio $\text{AUC}_{\text{generic}}/\text{AUC}_{\text{brand}}$ as a percent.
3. A generic is judged equivalent when this ratio falls within 80% to 125%. Does it pass?
4. The generic reaches a **higher peak**. Does that mean the body sees more drug?

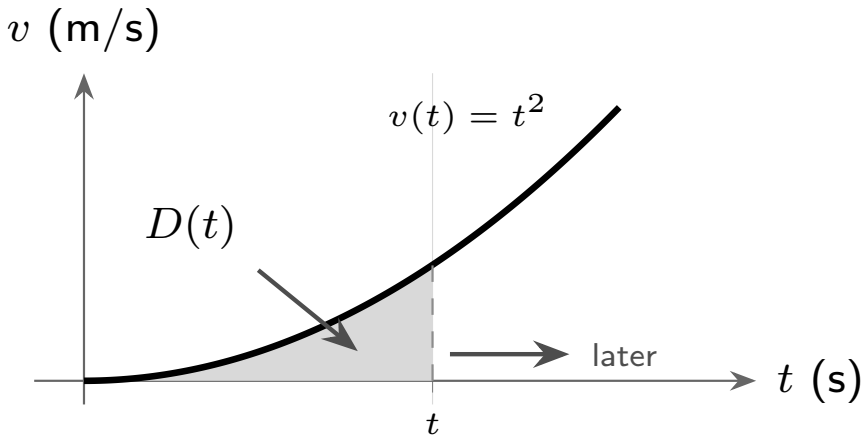
Exercise: Solution

Answers

1. With $\Delta t = 1$, each strip's area is just the average of its two ends:
   - Brand: $\frac{0+4}{2} + \frac{4+5}{2} + \frac{5+3}{2} + \frac{3+2}{2} = 2 + 4.5 + 4 + 2.5 = 13$ mg·h/L.
   - Generic: $\frac{0+6}{2} + \frac{6+4}{2} + \frac{4+2}{2} + \frac{2+1}{2} = 3 + 5 + 3 + 1.5 = 12.5$ mg·h/L.
2. $12.5/13 \approx 0.96$, so about 96%.
3. Yes: 96% sits inside 80% to 125%. (A real approval runs this on many volunteers with a statistical test, and compares peaks too.)
4. No. The generic peaks higher (6 vs. 5) and sooner, but its **total** exposure is slightly **lower**. The peak and the AUC answer different questions: how high, versus how much.

Antiderivatives and the Fundamental Theorem

A Function Out: The Distance So Far



Two kinds of process

- $\int_0^2 v(t)\,dt$: one fixed trip in, one **number** of meters out.
- A derivative: function in, **function** out. Can an integral do that too?

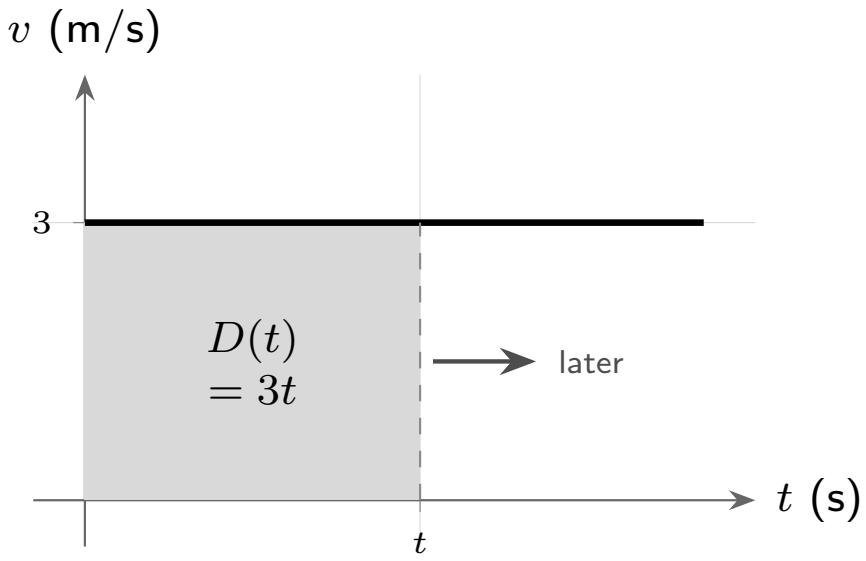
Let the stopping time move: the **distance so far**

$$D(t) = \int_0^t v(u)\,du$$

- Each stopping time t gives a distance, so D is a function: a trip odometer.
- u is the clock running up to t. It needs its own letter, since t is the stopping time.
- In general letters, this is the **area so far**: $A(x) = \int_a^x f(u)\,du$.

Notebook `U3-8_Integrals-2_FundamentalTheorem`, §1

Constant Velocity: The Distance So Far



$v(t) = 3$ m/s, from $t = 0$

- Distance so far: a rectangle, height 3, width t.

$$D(t) = 3t \text{ meters}$$

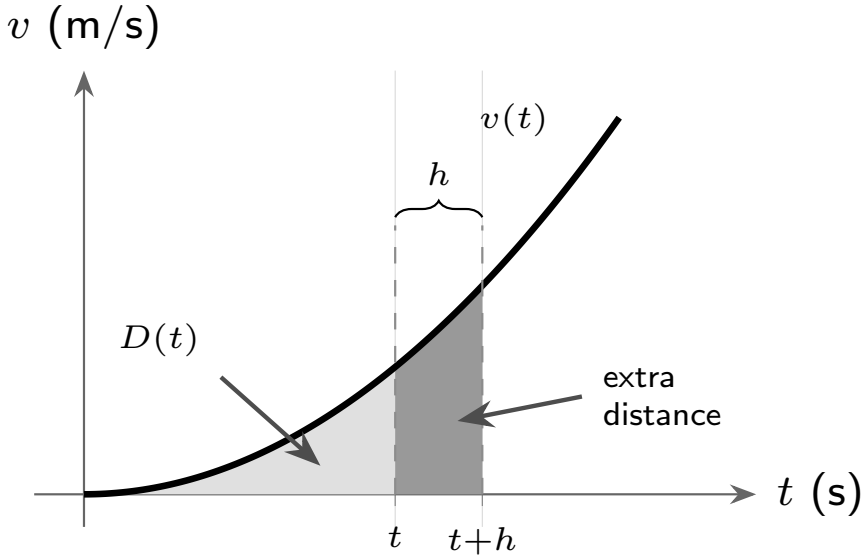
- Differentiate it:

$$D'(t) = 3 = v(t) \text{ m/s}$$

What just happened

- Differentiating the **distance so far** gave back the **velocity** we started with.
- Of course: the odometer climbs at the speedometer's rate. That is what velocity means.

Why $D'(t) = v(t)$: The Thin Sliver



Run the clock h seconds more

- The car goes a little farther: one thin **sliver**, $D(t+h) - D(t)$.
- Almost a rectangle: speed $v(t)$ times time h.

$$D(t+h) - D(t) \approx v(t)\, h$$

Divide by h, then let h shrink (Unit 3-1)

$$\frac{D(t+h) - D(t)}{h} \approx v(t) \quad \xrightarrow{h \to 0} \quad D'(t) = v(t)$$

In general letters: $A'(x) = f(x)$. The area so far grows at the rate of the height.

Notebook `U3-8_Integrals-2_FundamentalTheorem`, §2 and §3

Antiderivatives

Definition: antiderivative

D is an **antiderivative** of v if $D'(t) = v(t)$. It runs the derivative **backward**: "what distance function has this velocity?" In general letters: F is an antiderivative of f if $F'(x) = f(x)$.

Example: $v(t) = 2t$

- $D(t) = t^2$ works: $(t^2)' = 2t$.
- So do $t^2 + 1$ and $t^2 - 7$: a constant differentiates to 0.
- Every antiderivative is $t^2 + C$, where C **is the odometer's starting number**.

Notation

- The **indefinite integral** (no limits): $\int 2t\,dt = t^2 + C$.
- C is the **constant of integration**.
- **Always check**: differentiate, and you should get v back.

A Table of Antiderivatives

Each row reads backward from a derivative rule in Unit 3-1

$f(x)$	$\int f(x)\,dx$	**check: differentiate**
k (a constant)	$kx + C$	$(kx)' = k$
$x^n, \quad n \neq -1$	$\frac{x^{n+1}}{n+1} + C$	$\frac{(n+1)x^n}{n+1} = x^n$
$\frac{1}{x}$	$\ln\lvert x\rvert + C$	$(\ln x)' = \frac{1}{x}$
e^x	$e^x + C$	$(e^x)' = e^x$
$\cos x$	$\sin x + C$	$(\sin x)' = \cos x$
$\sin x$	$-\cos x + C$	$(-\cos x)' = \sin x$

x stands for whatever the input is. For the car, read x as t: $\int t^2\,dt = \frac{t^3}{3} + C$.

Rules

- Constants factor out: $\int kf = k\int f$.
- Sums split: $\int (f+g) = \int f + \int g$.

Watch out

$n = -1$ would divide by 0 in the power rule. That is why $\frac{1}{x} = x^{-1}$ gets its own row.

The Fundamental Theorem of Calculus

In the car

1. $D(t) = \int_0^t v(u)\,du$, the distance so far, has $D'(t) = v(t)$.
2. For **any** antiderivative D of v, the distance from time a to time b is

$$\int_a^b v(t)\,dt = D(b) - D(a)$$

In general letters

1. $A(x) = \int_a^x f(u)\,du$, the area so far, has $A'(x) = f(x)$.
2. For **any** antiderivative F of f, the area from a to b is

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Why any antiderivative works: read the odometer twice

- Distance traveled = reading at the end − reading at the start.
- Any antiderivative has a starting number C. Subtracting cancels it: use $C = 0$.
- So integration and differentiation **undo each other**.

Notebook `U3-8_Integrals-2_FundamentalTheorem`, §4

Back to the Car: Exactly $\frac{8}{3}$

$v(t) = t^2$, from $t = 0$ to 2 s

- Antiderivative: $D(t) = \frac{t^3}{3}$, since $D'(t) = t^2$.
- Fundamental Theorem:

$$\int_0^2 t^2\,dt = D(2) - D(0) = \tfrac{8}{3} - 0$$

- The car travels $\frac{8}{3} \approx 2.667$ m.

Where the sums were heading

n	**midpoint sum**
4	2.6250
8	2.6563
16	2.6641
100	2.6666
exact	$\frac{8}{3} = 2.6667$

And the signed-area example: the signs take care of themselves

For $v(t) = t - 1$, take $D(t) = \frac{t^2}{2} - t$. Then $D(3) - D(0) = \left(\frac{9}{2} - 3\right) - 0 = \frac{3}{2}$ m, matching $2 - \frac{1}{2}$ from the picture.

Exercise: Using the Fundamental Theorem

Try it

Find each integral exactly. Check your antiderivative by differentiating.

1. A cyclist's velocity is $v(t) = 6t - 3t^2$ m/s for $t = 0$ to 2 s. How far does the cyclist travel? ($v \geq 0$ on this interval.)
2. A patient's IV is being tapered off: the drip rate is $r(t) = 4 - t^2$ mL/min for $t = 0$ to 2 minutes. How much does the patient receive?
3. $\displaystyle\int_1^3 2x\,dx$. Then check the answer with geometry.
4. $\displaystyle\int_1^e \frac{1}{x}\,dx$

Exercise: Solution

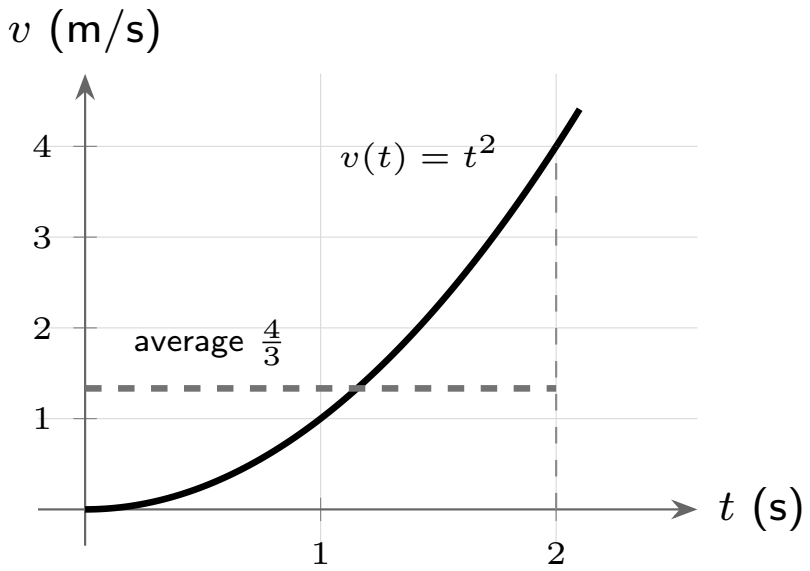
Answers

1. $D(t) = 3t^2 - t^3$ (check: $D'(t) = 6t - 3t^2$), so $D(2) - D(0) = (12 - 8) - 0 = 4$ m. Since $v \geq 0$, net and total distance agree.
2. $V(t) = 4t - \frac{t^3}{3}$ (check: $V'(t) = 4 - t^2$), so $V(2) - V(0) = \left(8 - \frac{8}{3}\right) - 0 = \frac{16}{3} \approx 5.33$ mL: rate (mL/min) times time (min) is a volume (mL).
3. $F(x) = x^2$, so $F(3) - F(1) = 9 - 1 = 8$. Geometry: a trapezoid with heights 2 and 6, width 2: $\frac{2+6}{2} \times 2 = 8$. ✓
4. $F(x) = \ln|x|$, so $F(e) - F(1) = 1 - 0 = 1$. The area under $\frac{1}{x}$ from 1 to e is exactly 1, one way to define the number e.

Applications of Integrals

The Average Value of a Function

So far, integrals turn a rate into a total. Two more jobs: finding an **average**, and finding a **probability**.



Under the dashed line and under the curve: the same area.

Definition: average value

$$\frac{1}{b-a}\int_a^b f(x)\,dx$$

The area, spread evenly across the width.

Worked: the car

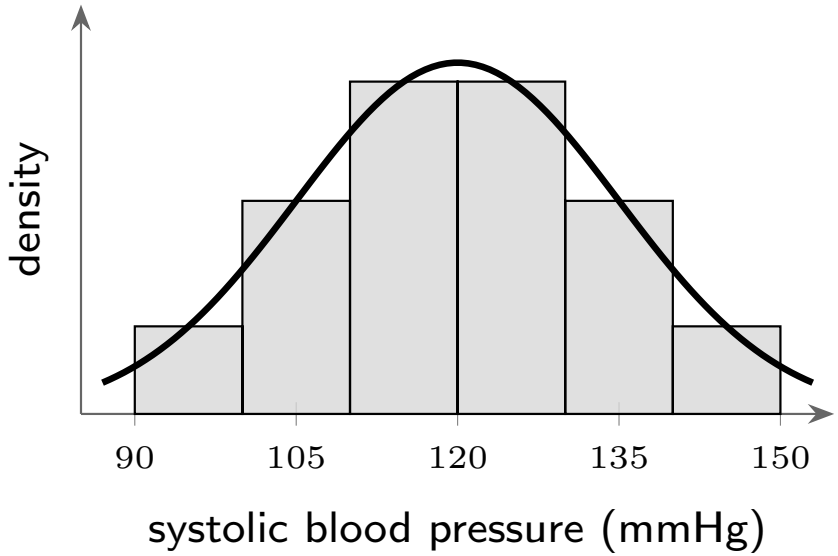
$$\frac{1}{2-0}\int_0^2 t^2\,dt = \frac{1}{2}\cdot\frac{8}{3} = \frac{4}{3}\ \text{m/s}$$

Total distance over total time.

Why we care

A patient's average drug level over a dosing interval is **AUC divided by the time**: mg·h/L over h is mg/L.

From Histogram to Density Curve



Bars become a curve

- A bar's **area** is the fraction of patients in that range.
- All the bars total 1.
- Narrower bins and more data smooth the bars into a curve.

Definition: probability density function

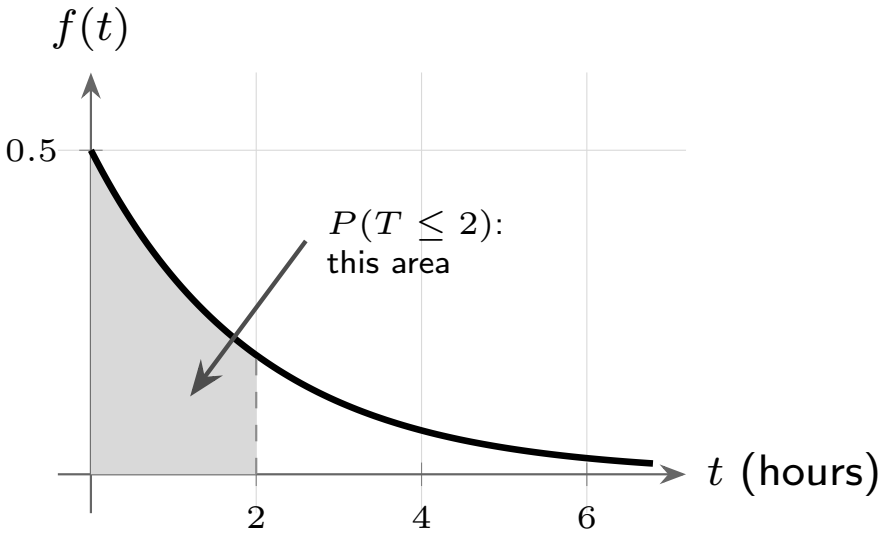
A **continuous random variable** X can land anywhere in a range. Its **density** $f(x)$ has $f(x) \geq 0$, and

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P(a \leq X \leq b) = \int_{a}^{b} f(x)\,dx$$

Probability is area under the density curve.

Notebook `DiscreteVsContinuous`, Unit 4

Worked: Waiting for the Next ER Arrival



T is the wait, with density $f(t) = 0.5\,e^{-0.5t}$ for $t \ge 0$: the **exponential** distribution.

The steps

1. Antiderivative: $G(t) = -e^{-0.5t}$.
   Check by the chain rule: $G'(t) = 0.5\,e^{-0.5t}$.
2. Total area: as t grows, $e^{-0.5t} \to 0$, so
   $0 - G(0) = 0 - (-1) = 1$. ✓
3. $P(T \le 2) = G(2) - G(0)$
   $= -e^{-1} + 1 \approx 0.632$.

What it means

There is about a 63% chance the next patient arrives within 2 hours.

Reading a Density: Probability So Far

Definition: cumulative distribution function (CDF)

The **probability so far**: $F(x) = P(X \le x) = \int_{-\infty}^{x} f(u)\,du$. For the ER wait, $F(t) = 1 - e^{-0.5t}$.

The same idea three times: the Fundamental Theorem

the rate	**the amount so far**	**its derivative**
velocity $v(t)$	distance so far $D(t)$	$D'(t) = v(t)$
height $f(x)$	area so far $A(x)$	$A'(x) = f(x)$
density $f(x)$	probability so far $F(x)$	$F'(x) = f(x)$

Watch out: a density is not a probability

- One exact value has **no width**, so no area: $P(X = c) = 0$. Ask about ranges.
- A density can exceed 1: height 2 from 0 to 0.5 still has total **area** 1.

Expected Value

Definition: expected value

The long-run average of X: each value x, weighted by its density, added up.

$$E[X] = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

The sum $\sum x_i\, P(x_i)$ for counted outcomes, with $\int$ in place of $\sum$ and $f(x)\,dx$ in place of $P(x_i)$.

Worked: uniform from 0 to 10

Height $\frac{1}{10}$, so the total area is 1.

Antiderivative of $x \cdot \frac{1}{10}$: $\quad G(x) = \frac{x^2}{20}$.

$$E[X] = G(10) - G(0) = \frac{100}{20} - 0 = 5$$

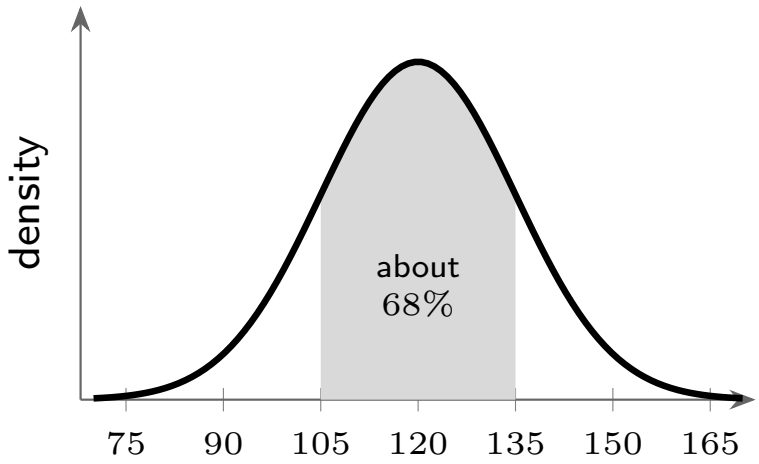
The middle, as it should be.

The ER wait

A density $\lambda e^{-\lambda t}$ has $E[T] = \frac{1}{\lambda}$.

With $\lambda = 0.5$ arrivals per hour, the average wait is $\frac{1}{0.5} = 2$ hours.

The Normal Distribution



Mean $\mu = 120$, standard deviation $\sigma = 15$: illustrative numbers.

The bell curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/(2\sigma^2)}$$

Unit 3-1's $e^{-x^2/2}$, moved and scaled so the area is 1. It is steepest at $\mu \pm \sigma$.

Area within $\mu \pm k\sigma$

$\pm 1\sigma$	$\pm 2\sigma$	$\pm 3\sigma$
68.3%	95.4%	99.7%

So $P(X > 150) \approx 2.3\%$: half of the 4.6% outside $\pm 2\sigma$.

Watch out: no formula for the area

This f has **no antiderivative** we can write down, so the Fundamental Theorem cannot help. Software adds up rectangles or trapezoids.

Exercise: Applications of Integrals

Try it

1. A drug level falls as $C(t) = 8 - 2t$ mg/L for $t = 0$ to 4 hours. Find the AUC and the average level, with units.
2. Let X have density $f(x) = \frac{x}{2}$ for x from 0 to 2, and 0 elsewhere.
   - Check that the total area is 1.
   - Find $P(X \le 1)$ and $P(1 \le X \le 2)$. The widths match, so why do these differ?
   - Find $E[X]$.
3. For the ER wait, $f(t) = 0.5\, e^{-0.5t}$, find the chance of waiting **more than** 1 hour.

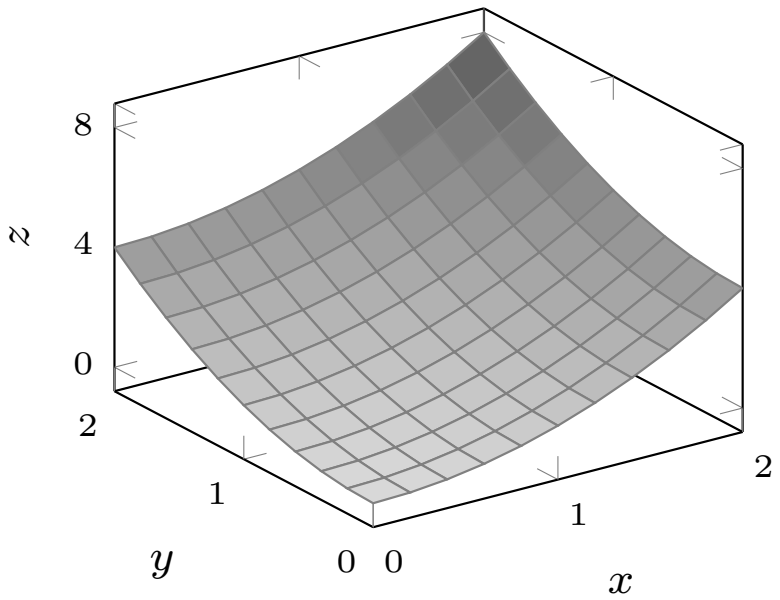
Exercise: Solution

Answers

1. $G(t) = 8t - t^2$, so AUC $= G(4) - G(0) = 32 - 16 = 16$ mg·h/L. Average: $\frac{16}{4} = 4$ mg/L.
2. With $G(x) = \frac{x^2}{4}$, whose derivative is $\frac{x}{2}$:
   - Total: $G(2) - G(0) = 1 - 0 = 1$. ✓
   - $P(X \leq 1) = G(1) - G(0) = \frac{1}{4}$, and $P(1 \leq X \leq 2) = G(2) - G(1) = \frac{3}{4}$. The density is **taller** on the right, so the same width holds more area.
   - $x \cdot \frac{x}{2} = \frac{x^2}{2}$ has antiderivative $\frac{x^3}{6}$, so $E[X] = \frac{8}{6} - 0 = \frac{4}{3}$. It sits right of the middle, 1, because more of the area does.
3. $G(t) = -e^{-0.5t}$ goes to 0 as t grows, so $P(T > 1) = 0 - G(1) = e^{-0.5} \approx 0.607$: about a 61% chance of waiting more than an hour.

2D Integrals

From Area to Volume



The bowl $z = x^2 + y^2$ (Unit 3-1), over x and y from 0 to 2.

The question

A function $f(x, y)$ has a **surface** for a graph. How much **volume** sits between it and a region R on the floor?

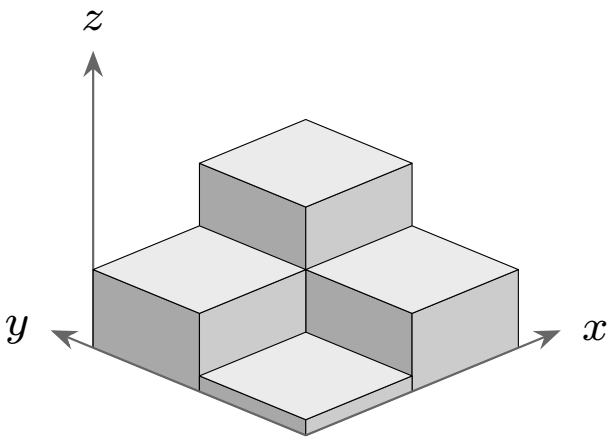
Where it shows up

- **Dose maps**: total radiation across a tissue slice.
- **Rainfall**: total water across a county.
- **Probability**: the chance two measurements both land in a region.

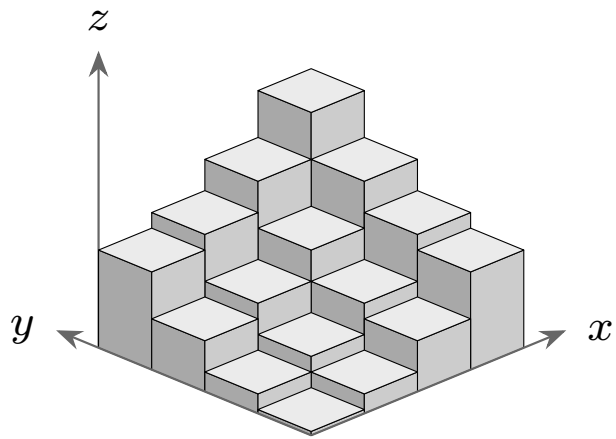
The plan that worked for area: chop, approximate with simple shapes, add up, let the pieces shrink.

Boxes Under the Surface

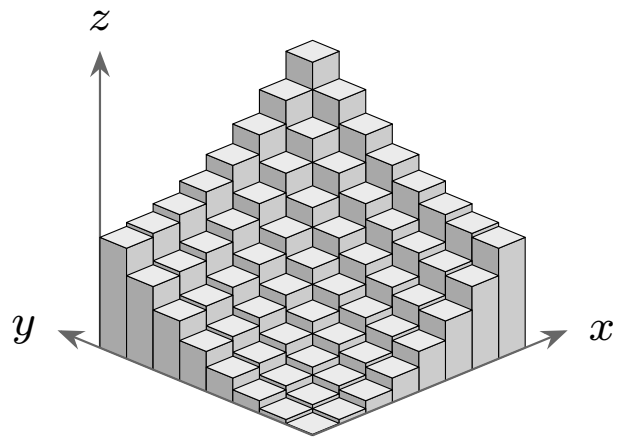
Grid the floor, and stand a **box** on each cell, as tall as the surface at the cell's middle.



2 per side



4 per side



8 per side

More boxes, closer answers: $f(x, y) = x^2 + y^2$, x and y from 0 to 2

per side	2	4	8	$\to \infty$
boxes	4	16	64	
volume	10	10.5	10.625	$\to \frac{32}{3} \approx 10.667$

Notebook `U3-8_Integrals-5_2D`, §2 and §3

The 2D Riemann Sum

Definition: 2D Riemann sum

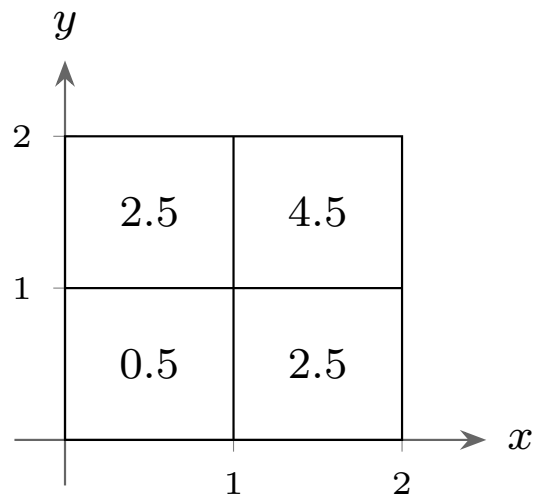
A box has volume height $\times$ width $\times$ depth, and we add up every box in the grid:

$$\sum_{i=1}^{n}\sum_{j=1}^{n} f(x_i^*, y_j^*)\,\Delta x\,\Delta y$$

Two sums: i walks across the x direction, and j across the y direction.

Worked: $x^2 + y^2$, 2 per side

1. Widths: $\Delta x = \Delta y = \frac{2-0}{2} = 1$.
2. Middles: x and y are each 0.5 or 1.5.
3. Heights: $0.5,\ 2.5,\ 2.5,\ 4.5$.
4. Volume: $(0.5 + 2.5 + 2.5 + 4.5) \times 1 \times 1 = 10$.



The floor from above: each cell's box height.

Notebook `U3-8_Integrals-5_2D`, §2

The Double Integral

Definition: double integral

Let the grid get infinitely fine: the sums settle on the **double integral** over R.

$$\lim_{n\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{n} f(x_i^*, y_j^*)\, \Delta x\, \Delta y \;=\; \iint_R f(x,y)\, dA$$

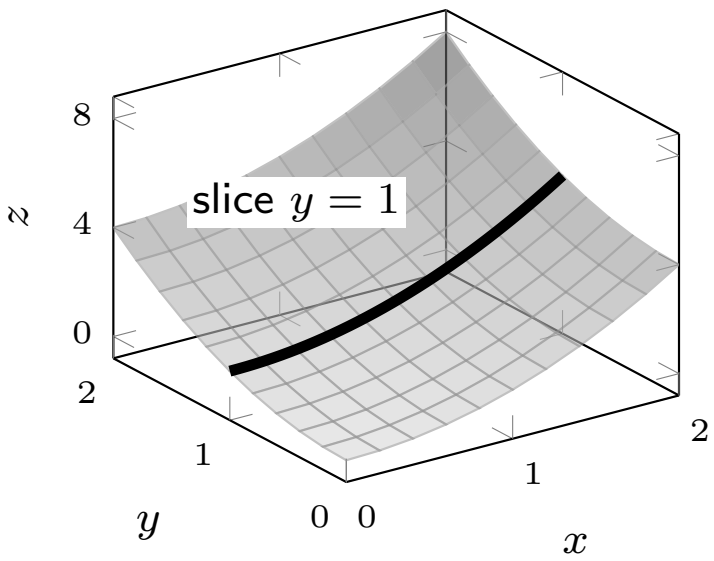
dA is a cell's floor area, $dx\, dy$, shrunk toward 0.

One variable and two, side by side

	one variable	**two variables**
graph	curve $y = f(x)$	surface $z = f(x,y)$
floor	interval from a to b	region R
piece	rectangle, $f\, \Delta x$	box, $f\, \Delta x\, \Delta y$
result	area, $\int_a^b f(x)\, dx$	volume, $\iint_R f(x,y)\, dA$
units	$f \times x$	$f \times x \times y$

Below the floor counts **negative**. A dose in mg/cm^2, over x and y in cm, integrates to a total in mg.

Exact Volume: One Variable at a Time



Freeze y, as with partial derivatives (Unit 3-1): each slice is a curve with an area.

Worked: $\displaystyle\int_0^2 \left(\int_0^2 (x^2 + y^2)\, dx \right) dy$

Inside first. Treat y as a constant. An antiderivative in x is $\frac{x^3}{3} + y^2 x$, so from $x = 0$ to 2 the slice has area

$$S(y) = \tfrac{8}{3} + 2y^2.$$

Then add up the slices. An antiderivative in y is $\frac{8}{3}y + \frac{2}{3}y^3$, so from $y = 0$ to 2:

$$\tfrac{16}{3} + \tfrac{16}{3} = \tfrac{32}{3} \approx 10.667.$$

What it says

This is the number the boxes were closing in on. For a smooth f over a rectangle, either order gives the same answer.

Notebook `U3-8_Integrals-5_2D`, §1

The Cost of a Dimension

A grid with 100 cuts per direction

inputs	**evaluations**	**at 10^6 per sec**
1	100	instant
2	$10{,}000$	instant
3	10^6	1 second
4	10^8	under 2 minutes
7	10^{14}	about 3 years

The curse of dimensionality

- n cuts in d dimensions means n^d evaluations.
- Each new input **multiplies** the work.

Why we care

- Real data has many inputs, so no one integrates a 20-dimensional function on a grid.
- The way out is to drop the grid and **sample at random**: Monte Carlo integration (Unit 4).

Notebook `U3-8_Integrals-5_2D`, §4

Exercise: 2D Integrals

Try it

1. Let $f(x, y) = 3$ over x from 0 to 2 and y from 0 to 5. What shape is under the surface? Find its volume with no calculus.
2. Let $f(x, y) = x + y$ over x and y from 0 to 2. Estimate the volume with 2 boxes per side, using each cell's middle.
3. Find the same volume exactly, one variable at a time, x first.
4. $f(x, y)$ is a radiation dose in mg/cm^2, with x and y in cm. What are the units of $\iint_R f(x, y)\, dA$, and what does the number mean?

Exercise: Solution

Answers

1. A flat roof at height 3 makes one big box: $3 \times 2 \times 5 = 30$.
2. $\Delta x = \Delta y = 1$, and the middles are 0.5 and 1.5. Heights: $1,\ 2,\ 2,\ 3$.

   Volume: $(1 + 2 + 2 + 3) \times 1 \times 1 = 8$.
3. Treat y as a constant. An antiderivative in x is $\frac{x^2}{2} + yx$, so from $x = 0$ to 2 the slice has area $S(y) = 2 + 2y$.

   An antiderivative in y is $2y + y^2$, so from $y = 0$ to 2: $4 + 4 = 8$.

   The estimate was exact: this surface is a flat, tilted plane, so each box's extra on one side cancels its shortfall on the other.
4. Units: mg/cm$^2 \times$ cm $\times$ cm $=$ mg. The number is the **total dose** delivered across the region R.

Review: Integrals

What we covered: Riemann sums and the definite integral

1. A rate times a time is an amount. When the rate changes, the amount is the **area under the rate's graph**: distance from velocity, exposure from concentration.
2. A **Riemann sum** cuts $[a, b]$ into n strips of width $\Delta x = \frac{b-a}{n}$ and adds up rectangles: $\sum_{i=1}^{n} f(x_i^*)\,\Delta x$. Heights from the left end, right end, or middle. For an increasing f, left is too low and right too high; midpoint is usually best.
3. The **definite integral** is the limit as $n \to \infty$: $\int_a^b f(x)\,dx$. $\int$ is the sum, $f(x)$ the height, dx the width.
4. Area below the axis counts **negative**, so the integral is a **net** amount. Its units are (units of f) $\times$ (units of x).

Review: Integrals

What we covered: data and the Fundamental Theorem

5. With **data**, the sampling schedule fixes Δx. The **trapezoid rule**, the average of left and right, is the standard choice; it is how AUC is computed. In code, `np.trapezoid(y, x)`.
6. The **distance so far**, $D(t) = \int_0^t v(u)\,du$, has $D'(t) = v(t)$: the thin-sliver argument. In general letters, the area so far has $A'(x) = f(x)$.
7. An **antiderivative** runs the derivative backward: $D' = v$, or $F' = f$. It comes with a $+C$, the odometer's starting number. Check one by differentiating.
8. **Fundamental Theorem:** $\int_a^b v(t)\,dt = D(b) - D(a)$, and in general $\int_a^b f(x)\,dx = F(b) - F(a)$. Integration and differentiation undo each other, and exact areas need no rectangles.

Review: Integrals

What we covered: applications

9. The **average value** of f is $\frac{1}{b-a}\int_a^b f(x)\,dx$: the area, spread evenly across the width. Average drug level is AUC over time.
10. A **density** $f(x)$ has $f(x) \geq 0$ and total area 1. **Probability is area**: $P(a \leq X \leq b) = \int_a^b f(x)\,dx$, and any one exact value has probability 0.
11. The **CDF** $F(x) = P(X \leq x)$ is the probability so far, with $F'(x) = f(x)$. The **expected value** is $E[X] = \int x\, f(x)\,dx$.
12. The **normal** density has no antiderivative we can write down, so its areas come from software: about 68%, 95%, 99.7% within 1, 2, 3 standard deviations.

Review: Integrals

What we covered: 2D integrals

13. Over a region R, rectangles become **boxes** and area becomes **volume**. The **2D Riemann sum** is $\sum_i \sum_j f(x_i^*, y_j^*)\,\Delta x\,\Delta y$.
14. The **double integral** $\iint_R f(x,y)\,dA$ is its limit as the grid gets infinitely fine. Its units are (units of f) $\times$ (units of x) $\times$ (units of y).
15. Exact volumes go **one variable at a time**: freeze y and integrate in x to get a slice's area, then integrate the slices in y.
16. A grid needs n^d evaluations in d dimensions: the **curse of dimensionality**. Beyond a few inputs, sample at random instead.

Notebooks: `U3-8_Integrals-1` to `-5`; `DiscreteVsContinuous`. **Previous:** Unit 3-2, Optimization and Linear Regression. This closes Unit 3, **Calculus**.